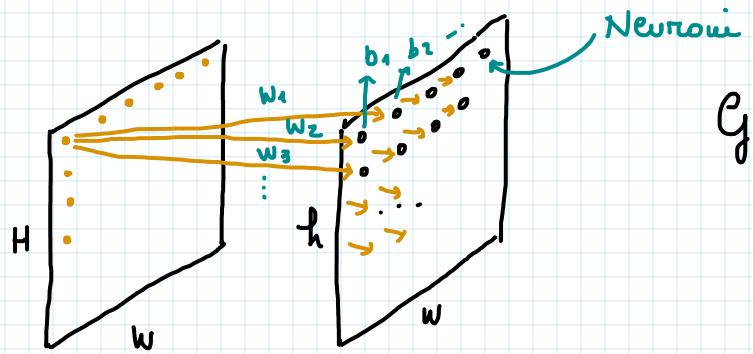
CNN

Def Si dice CORRELAZIONE INCROCIATA (CROSS-CORRELATION) di un segnale in INPUT $x\ \mathbb{R}\to\mathbb{R}$ contro $w:\mathbb{R}\to\mathbb{R}$ (KERNEL):

$$(x \star w)(t) = \int_{\mathbb{R}} x(a)\, w(t+a)\, da = (x * \underbrace{w^-}_{\text{KERNEL FLIPPATO}})(-t)$$

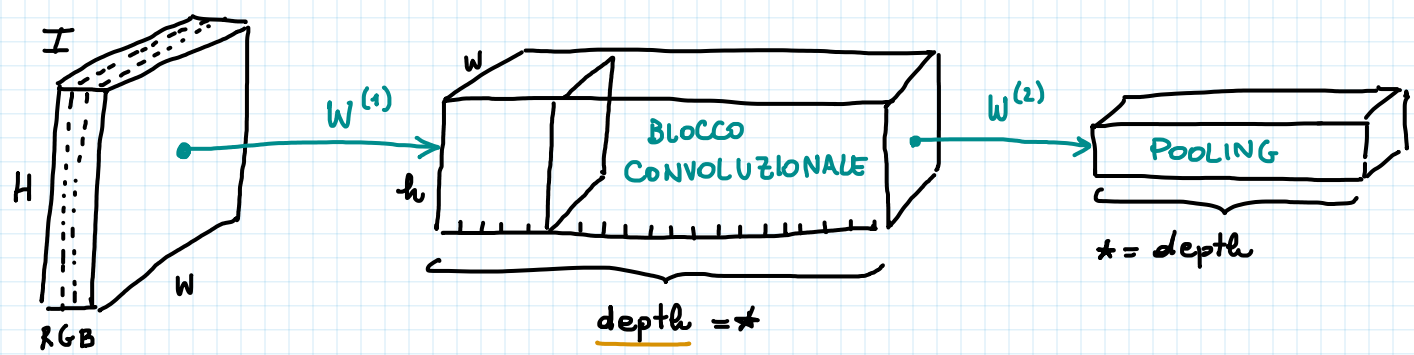
KERNEL FLIPPATO: $w^-(t) = w(-t)$

RETI NEURALI CONVOLUZIONALI

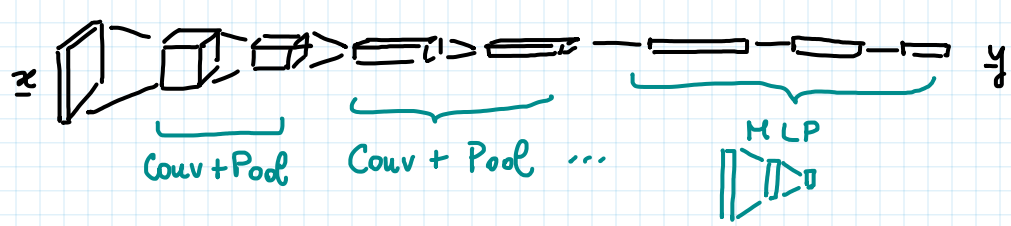


Gestire dati a "GRIGLIA"

Ragionamento a VOLUMI $\rightsquigarrow$ TENSORI



Struttura o ARCHITETTURA standard di una CNN:



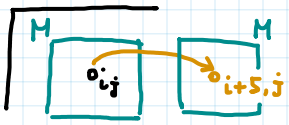
INPUT

$$I \in \mathbb{R}^{H\times W\times \#channel} \xrightarrow{W^{(1)},\, b^{(1)}} C \in \mathbb{R}^{h_1\times w_1\times \#filtri} \xrightarrow{W^{(2)},\, b^{(2)}} P \in \mathbb{R}^{h_2\times w_2\times \#filtri}$$

LAYER / BLOCCO CONVOLUZIONALE

Def. Gli IPERPARAMETRI di un blocco CONVOLUZIONALE sono:

- DEPTH : profondità , # filtri $\boxed{d}$
- STRIDE : # pixel di cui "SPOSTIAMO" il filtro $\boxed{s}$

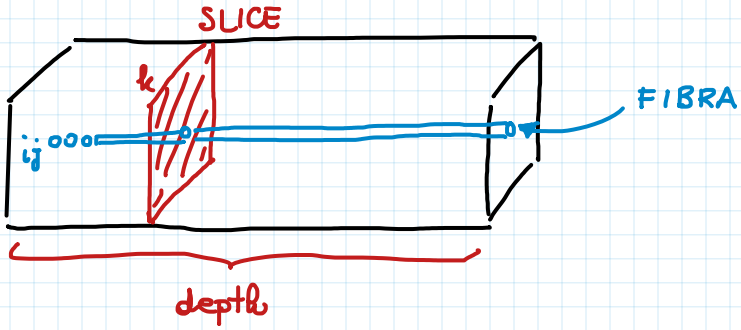


- ZERO-PADDING : dimensione del "CONTORNO" di ZERI attorno all'input $\boxed{p}$
- DIMENSIONE FILTRI / KERNEL : $m \times m$ con $\boxed{m}$ dispari

$$h_1 \times w_1 = \frac{H \times W - m + 2p}{s} + 1$$

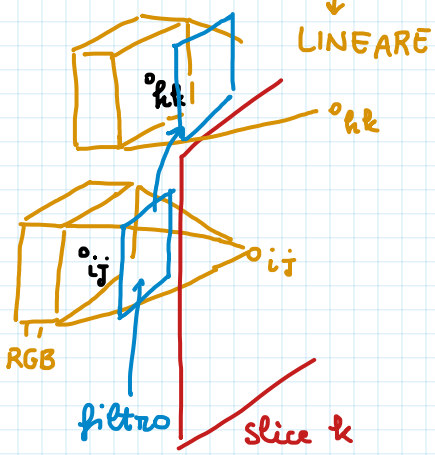
Def : SLICE

FIBRA



Qual è l'OUTPUT del BLOCCO CONVOLUZIONALE

$$c^k_{ij} = \sigma\left(\underbrace{W^{(1)}_k * \underline{x}}_{\text{LINEARE}} + \underline{b}^{(1)}\right) = \sigma\left(\sum_c \sum_\ell \sum_m \overset{\text{slice } k}{w^{(1)}_{\ell m}} \cdot x^{(c)}_{i-\ell,\, j-m} + \underline{b}^{(1)}\right)$$



- PESI CONDIVISI con tutti i pixel

filtro $\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ \vdots & & \\ & & w_{33} \end{bmatrix}$

Quindi se $m = 3$, ci sono SOLO 9 PARAMETRI da addestrare PER k

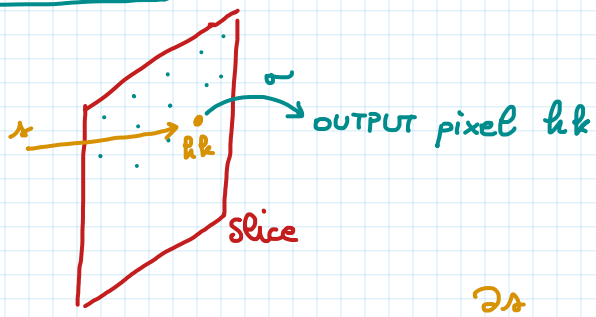
BLOCCO CONVOLUZIONALE · BACKPROPAGATION

- LOSS L_2 (MSE) : $\mathcal{L}$
- $$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(\ell)}} = \delta_j^{(\ell)} \cdot x_i$$

$$\delta_j^{(\ell)} = \sum_{v=1}^{m_\ell} \delta_v^{(\ell+1)} \, w_{jv}^{(\ell+1)} \cdot \boxed{\frac{d\sigma}{d s_j^{(\ell)}} \left(s_j^{(\ell)} \right)}$$

qui c'è la CONVOLUZIONE

$$\underbrace{s_{\cancel{j}\,hk}^{(\ell)} = \left(W^{(1)} * \underline{x} + \underline{b}^{(1)} \right)_{hk}}$$



$$\frac{d\sigma}{d s_{hk}^{(\ell)}} \left(s_{hk}^{(\ell)} \right) = \sum_h \sum_k \frac{d\sigma}{ds} \left(W^{(1)} * \underline{x} + \underline{b}^{(1)} \right)_{hk} \cdot \overbrace{x_{i-h,\, j-k}^{(c)}}^{\frac{\partial s}{\partial w_{ij}}} =$$

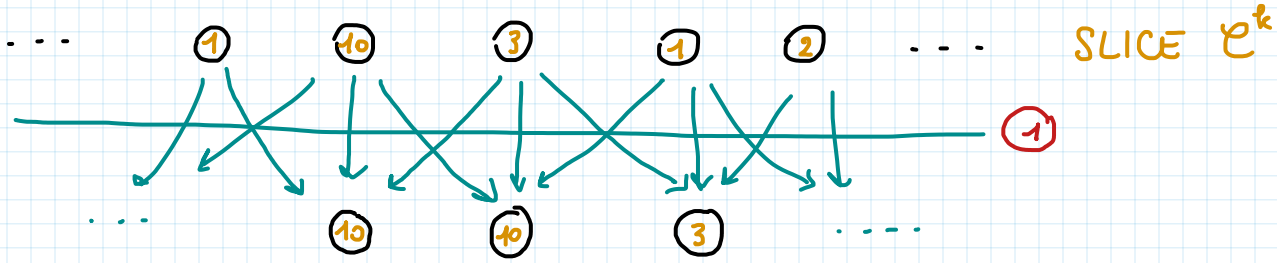
$$= \frac{d\sigma}{ds} \left(W^{(1)} * \underline{x} + b^{(1)} \right) * \underline{x}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial w_{ij}^{(\ell)}} = \sum_{v=1}^{m_\ell} \delta_v^{(\ell+1)} \, w_{jv}^{(\ell+1)} \cdot \left(\frac{d\sigma}{ds} * \underline{x} \right)_i}$$

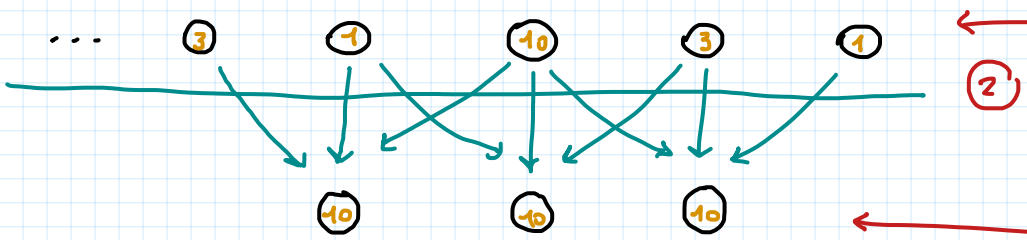
viene implementata come CROSS-CORRELATION

LAYER di POOLING ⇝ FILTRO NON-LINEARE

Standard : MAX-POOLING



SLICE $\mathcal{C}^k$



tutti sono cambiati rispetto CASO ①.

(2)

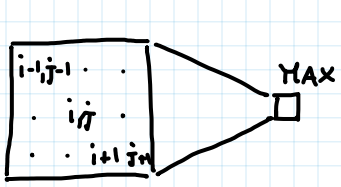
Due neuroni sono uguali al caso ①.

Def: IPERPARAMETRI del layer di POOLING sono:

DIMENSIONE FILTRO/KERNEL : $m \times m$ con $\boxed{m}$ dispari

STRIDE: $\boxed{s}$

$$x_{ij} \longmapsto \max\left(x_{i+m,\, j+m}\right) = p_{ij}^{k}$$



PROBLEMA: BACKPROPAGATION

La fz MAX NON è differenziabile

→ i valori di INPUT NON massimi ANNULLANO $\nabla \mathcal{L}$

→ il valore p_{ij}^{k} (che è il MASSIMO) viene riportato così com'è: $x \longmapsto x$

$$f'(x) = 1$$

abbiamo solo s_j

$$\delta_j^{(\ell)} = \sum_{v=1}^{M_\ell} \delta_v^{(\ell+1)} w_{jv}^{(\ell+1)} \frac{d\cancel{\sigma}}{d s_j}(s_j) = \sum_{v=1}^{M_\ell} \delta_v^{(\ell+1)} w_{jv}^{(\ell+1)} \underbrace{\mathbb{1}_{\{s_j = p_{ij}^{k}\}}}_{\text{è } =1 \text{ solo se } s_j \text{ è MASSIMO}}$$

$f: x \mapsto x$ se x è il massimo

RETI NEURALI RICORRENTI

MLP $\longrightarrow$ INPUT $\underline{x} \in \mathbb{R}^N$, N finito $\quad x$

CNN $\longrightarrow$ INPUT $I \in \mathbb{R}^{H \times W \times c}$ dove H, W, c sono finiti $\quad I$

DATO SEQUENZIALE

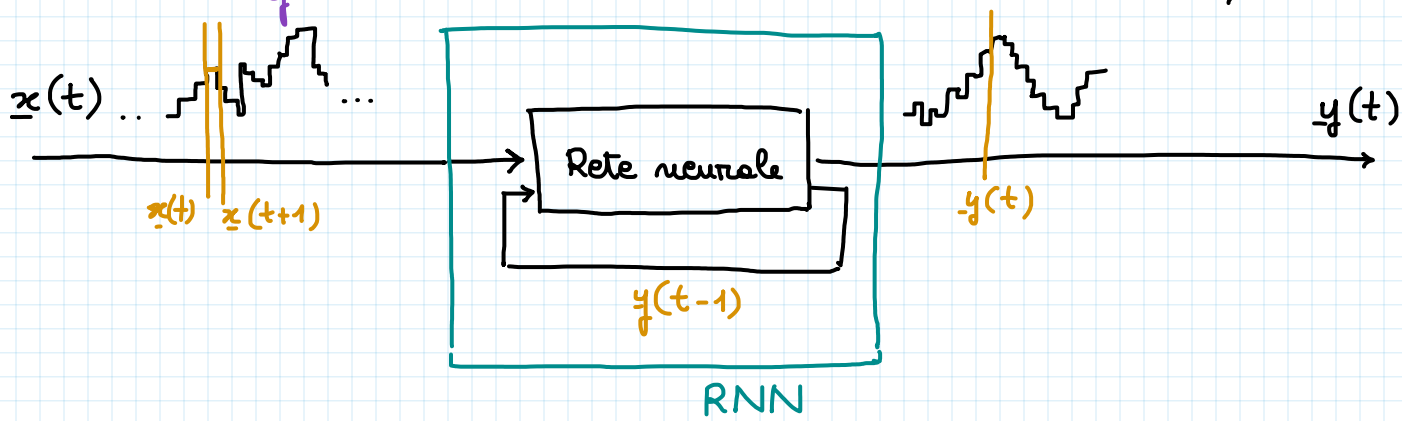
↳ INPUT : musica, video, serie temporali, testo

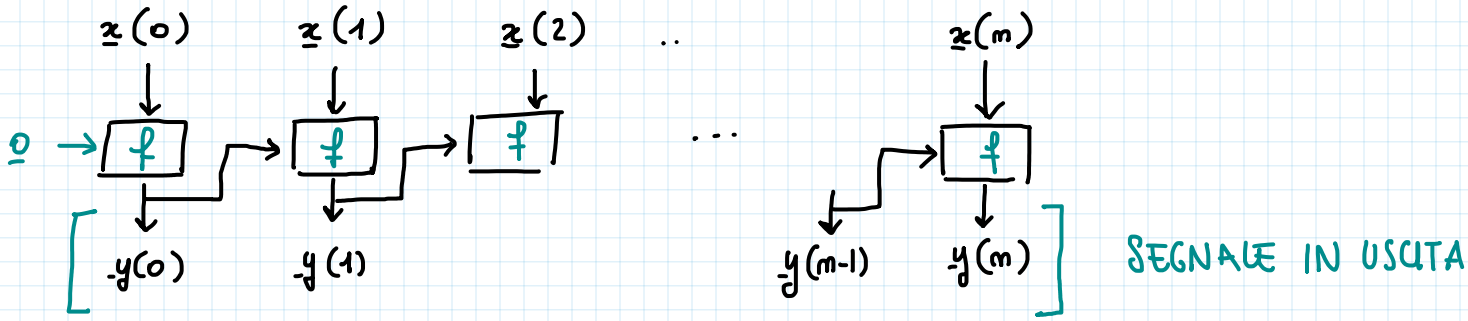
↳ OUTPUT : • speech synthesis IN/OUT

• image captioning OUT

• generazione di testo

PROBLEMA: Gestire la DIMENSIONE VARIABILE t ↪ CICLI





$$\underbrace{\sigma\left(W \cdot \underline{x}(t) + \hat{W}\, \underline{y}(t-1) + \underline{b}\right)}$$

$$\sigma\left(W^2 \cdot \boxed{} + \underline{b}^2\right) \quad \ldots$$

$$\sigma\left(W \cdot \underline{x}(t+1) + \hat{W}\, \underline{y}(t) + \underline{b}\right)$$

A t finito

$\left.\begin{matrix} \underline{x} \\ \underline{y} \end{matrix}\right\} \in \mathbb{R}^N$ dim fissa